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DEPARTMENT OF ELECTRONICS ENGINEERING

PROJECT SYNOPSIS

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STM32-Based Multisensor Industrial Condition Monitoring, Anomaly Detection, and Predictive Maintenance Using TinyML

Under the Guidance of

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ABSTRACT

Industrial machines can develop abnormal operating conditions due to changes in temperature, vibration, electrical consumption, and rotational speed. This project proposes an STM32-Based Multisensor Industrial Condition Monitoring, Anomaly Detection, and Predictive Maintenance Using TinyML support.

The proposed system will use the STM32 NUCLEO-F446RE as the main controller, interfaced with temperature, vibration, electrical, and rotational sensors and an N20 geared motor for laboratory testing. A 2.8-inch TFT display, buzzer, and visual indicators will provide monitoring information and alerts. CAN and RS-485 interfaces will also be explored for communication with external devices.

The work will include sensor integration, data acquisition, signal processing, and threshold-based monitoring, followed by dataset preparation, model development, and deployment of a suitable lightweight machine-learning model on the STM32. The system will be evaluated in terms of anomaly-detection performance, memory usage, inference time, and overall operation. The expected outcome is a laboratory-scale prototype that combines conventional monitoring with on-device anomaly detection to provide condition information for predictive-maintenance decisions.

1. INTRODUCTION

Industrial equipment such as motors, pumps, and fans operates under changing mechanical, electrical, and thermal conditions. Parameters such as vibration, temperature, current, voltage, and rotational speed can provide useful information about equipment health. Conventional condition-monitoring systems often use predefined thresholds to identify abnormal conditions. However, fixed thresholds may not adequately identify changes that involve multiple parameters or develop under different operating conditions.

Advances in microcontrollers have made it possible to acquire and process sensor data locally. Edge AI extends this capability by allowing trained machine-learning models to analyse data directly on embedded devices, reducing dependence on external computing systems. However, the limited memory and processing capacity of microcontrollers require suitable models to be selected and evaluated carefully.

This project proposes an **STM32-Based Multisensor Industrial Condition Monitoring, Anomaly Detection, and Predictive Maintenance Using TinyML**. The STM32 NUCLEO-F446RE will serve as the main controller, while an N20 geared motor will provide a laboratory-scale test setup. The system will combine multisensor data acquisition, conventional threshold-based monitoring, and a lightweight machine-learning model for on-device anomaly detection.

The proposed work will focus on identifying selected abnormal operating conditions and presenting the results through a local display and alert mechanism. CAN and RS-485 com-

munication will also be explored for data exchange with external devices. The system will be evaluated to determine the suitability of lightweight Edge AI for embedded condition monitoring and its potential to support predictive-maintenance decisions within the resource limitations of the STM32 platform.

2. PROBLEM STATEMENT

Industrial machines may develop abnormal operating conditions due to changes in vibration, temperature, electrical load, rotational speed, or other developing faults. Conventional monitoring systems often rely on individual sensor readings and fixed threshold values. Although these methods can identify clear deviations from normal operating limits, they may not adequately detect abnormal patterns involving multiple parameters.

Machine-learning techniques can help analyse such patterns, but running them on embedded hardware requires suitable models that can operate within limited memory and processing capacity. Using external computers or cloud platforms for analysis may also increase system complexity and dependence on external resources.

Therefore, this project proposes an **STM32 NUCLEO-F446RE-based multisensor condition-monitoring system** that combines local sensor-data processing, threshold-based monitoring, and lightweight Edge AI for anomaly detection. The system will integrate temperature, vibration, electrical, and rotational measurements, along with a local display, alerts, and CAN/RS-485 communication interfaces.

The main problem addressed is whether a lightweight machine-learning model can be deployed on the STM32 to identify selected abnormal operating conditions alongside conventional monitoring. The prototype will be evaluated for anomaly-detection performance, resource requirements, and overall operation, with the aim of providing condition information to support predictive-maintenance decisions.

3. OBJECTIVES

The main objective of this project is to develop an **STM32-Based Multisensor Industrial Condition Monitoring, Anomaly Detection, and Predictive Maintenance Using TinyML**. The specific objectives are:

- To develop an embedded monitoring system using the STM32 NUCLEO-F446RE for a laboratory-scale motor condition-monitoring setup.
- To interface temperature, vibration, electrical, and rotational sensors and acquire the corresponding data through embedded firmware.
- To implement local data processing, feature extraction, threshold-based monitoring, display of operating conditions, and abnormal-condition alerts.

- To collect and analyse sensor data under normal and selected abnormal operating conditions and develop a suitable lightweight machine-learning model for on-device anomaly detection.
- To deploy and evaluate the selected model on the STM32, considering detection performance, memory usage, and inference time.
- To explore CAN and RS-485 communication and evaluate the integrated prototype for condition monitoring and predictive-maintenance support under laboratory conditions.

4. EXISTING SYSTEM / LITERATURE OVERVIEW

Industrial condition monitoring uses parameters such as vibration, temperature, current, voltage, and rotational speed to assess the operating condition of machinery. Chang *et al.* [1] developed a motor condition-monitoring system using a Cortex-M4 microcontroller to process electrical and vibration signals. Their work demonstrates how sensor acquisition and signal processing can be performed on embedded hardware for motor abnormality detection.

Surucu *et al.* [2] reviewed machine-learning techniques for condition monitoring and discussed the importance of selecting suitable data-processing methods and models for reliable fault detection. Machine learning can help identify patterns in sensor data, but its effectiveness depends on the quality of the data and the suitability of the selected model.

Artiushenko *et al.* [3] investigated a resource-efficient Edge AI system for tool condition monitoring using machine-learning and deep-learning models on an edge device. Arciniegas *et al.* [4] studied motor vibration monitoring using TinyML, combining vibration analysis and deep learning on a resource-constrained device. These studies provide relevant examples of local data analysis and machine-learning deployment for condition monitoring.

Based on this work, the proposed project will combine multisensor data acquisition, conventional threshold-based monitoring, and lightweight Edge AI on the STM32 NUCLEO-F446RE. The system will use temperature, vibration, electrical, and rotational measurements for local condition assessment. A suitable model will be developed and evaluated for on-device anomaly detection, while CAN and RS-485 communication will be explored for data exchange. The prototype will be evaluated under laboratory conditions, considering detection performance, memory usage, and inference time.

5. PROPOSED SYSTEM / METHODOLOGY

The proposed system will use the STM32 NUCLEO-F446RE as the main controller for multisensor industrial condition monitoring and Edge AI-based anomaly detection. An N20 geared motor, controlled through an L298N motor driver, will provide a laboratory-scale test setup.

The system will acquire temperature, vibration, electrical, and rotational data using the DS18B20, MPU6050, INA219, and motor encoder/A3144 Hall-effect sensor. The embedded

firmware will handle data acquisition, signal processing, threshold-based monitoring, and communication. A 2.8-inch TFT display, passive buzzer, and visual indicators will provide operating information and abnormal-condition alerts.

The proposed system workflow, including data processing, model development and deployment, anomaly detection, communication, and system evaluation, is shown in **Figure 1**.

The functional flow is:

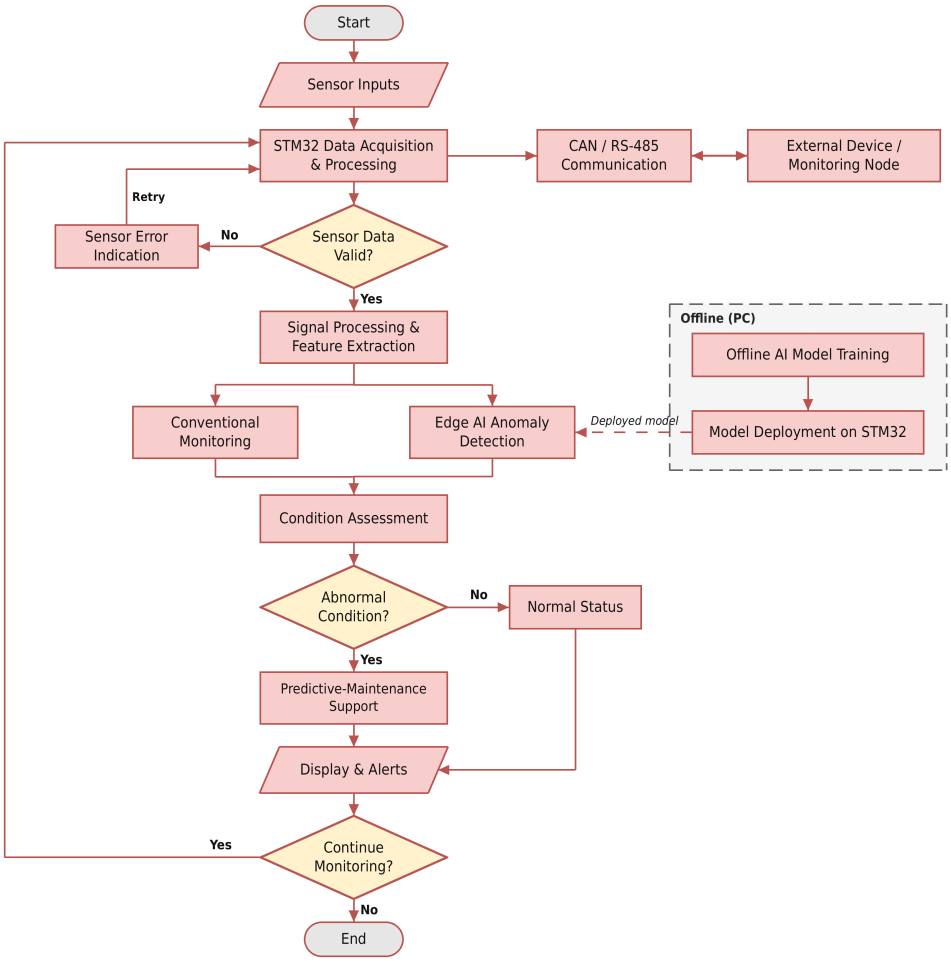


Figure 1. Proposed System Flowchart

Initially, threshold-based monitoring will be used to verify sensor operation and establish baseline conditions. Data collected under normal and selected abnormal operating conditions will be analysed on a computer for feature preparation and model development.

Suitable lightweight machine-learning models, such as an MLP or 1D-CNN, will be evaluated according to the collected data and STM32 resource constraints. The selected model will be optimised and deployed on the STM32 for local inference. Evaluation will consider anomaly-detection performance, memory usage, computational requirements, and inference time.

CAN and RS-485 communication will be explored for data exchange with external devices or monitoring nodes. The integrated system will combine conventional monitoring with

Edge AI-based analysis to identify selected abnormal conditions and provide information for predictive-maintenance decisions.

6. HARDWARE AND SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

Component	Purpose
STM32 NUCLEO-F446RE	Main controller, data processing and Edge AI inference
MPU6050	Vibration and motion measurement
DS18B20	Temperature measurement
INA219	Voltage, current and power measurement
N20 6 V, 100 RPM motor with encoder	Motor test setup and rotational feedback
A3144 Hall-effect sensor	Additional speed sensing
L298N motor driver	Motor control
SN65HVD230 CAN modules	CAN communication
MAX3485 RS-485 module	RS-485 communication
2.8-inch SPI TFT display	Main monitoring interface
SSD1306 OLED	Secondary status display, if required
Passive buzzer and visual indicators	Fault alerts and status indication
7-segment displays and relay module	Numerical indication and output switching, if required
LM2596 buck converter	DC power regulation
Resistors and capacitors	Bus termination, pull-ups and supply decoupling
Push buttons, breadboards, headers and wires	Prototype assembly and control
USB cable	Programming, debugging and serial communication

The selected components will support laboratory-scale monitoring of mechanical, thermal, electrical and rotational conditions. The motor will provide a controllable setup for collecting sensor data under different operating conditions.

6.2 Software Requirements

Software/Tool	Purpose
STM32CubeIDE / STM32CubeMX	Firmware development, peripheral configuration and debugging
Embedded C	STM32 firmware implementation
Python	Data logging, processing, visualization and feature preparation
TensorFlow/Keras or a suitable ML framework	Model development and evaluation
STM32-compatible Edge AI framework/library	Model conversion and on-device inference
Serial terminal and data-logging tools	Debugging and experimental data collection
PuTTY	Viewing the serial output using the selected COM port and baud rate

The final machine-learning model will be selected based on experimental data, anomaly-detection performance, and the memory and computational constraints of the STM32F446RE. The required model-conversion and deployment tools will depend on the selected model and its compatibility with the target hardware.

7. IMPLEMENTATION PLAN AND SYSTEM EVALUATION

The project will be implemented in stages, with individual hardware and software modules verified before complete system integration.

Phase	Major Activities
STM32 Firmware	GPIO, timers, interrupts, UART, I2C, SPI and ADC
Sensor & Motor Integration	Sensor interfacing, motor control and speed measurement
Monitoring & Communication	Data processing, display, alerts, CAN and RS-485
AI Development	Dataset collection, preprocessing, model training and evaluation
Edge AI Deployment	Model optimization and deployment on STM32
Integration & Testing	System integration and experimental validation

The integrated prototype will be evaluated for sensor-data reliability, communication performance, anomaly-detection performance, RAM and Flash usage, inference time, and overall responsiveness. Quantitative results and the detectable abnormal conditions will be established through experiments under laboratory conditions.

8. EXPECTED OUTCOMES, APPLICATIONS AND LIMITATIONS

8.1 Expected Outcomes

The project is expected to deliver a laboratory-scale STM32-based condition-monitoring prototype capable of:

- Acquiring and processing temperature, vibration, electrical, and rotational data.
- Monitoring operating conditions using conventional threshold-based methods.
- Deploying a suitable lightweight Edge AI model for on-device anomaly detection.
- Displaying sensor readings and generating alerts for abnormal conditions.
- Supporting CAN and RS-485 communication for data exchange with external devices.
- Evaluating anomaly-detection performance, memory usage, and inference time.
- Providing condition information to support predictive-maintenance decisions.

Actual detection performance and resource requirements will be established through experimental testing.

8.2 Applications

The proposed system may serve as a foundation for condition monitoring of electric motors, fans, pumps, and other rotating machinery. It may also support educational and research activities in embedded systems, TinyML, and industrial monitoring. The initial implementation will focus on a laboratory-scale motor setup, and further testing under real operating conditions will be required before considering industrial deployment.

8.3 Limitations

The main limitations include the STM32's memory and processing capacity, the quality and diversity of collected data, sensor measurement accuracy, and the limited representation of industrial conditions in the laboratory setup. Therefore, anomaly-detection performance and the usefulness of the system for predictive maintenance will depend on experimental validation and the range of operating conditions represented in the dataset.

9. FUTURE SCOPE AND CONCLUSION

9.1 Future Scope

The system may be extended by integrating additional sensors, collecting larger datasets, and evaluating improved Edge AI models. Future work could include fault classification,

equipment degradation analysis, and remaining-useful-life estimation. Distributed monitoring through CAN or RS-485, a dedicated PCB, improved power management and protection, and a gateway for remote monitoring may also be explored. Further testing on actual industrial motors, pumps, and fans will be necessary to assess performance under real operating conditions.

9.2 Conclusion

This project aims to develop an STM32-based lightweight Edge AI system for multisensor industrial condition monitoring, anomaly detection, and predictive maintenance. The proposed system will combine sensor data acquisition, embedded processing, conventional monitoring, industrial communication, and on-device machine learning using the STM32F446RE.

The work will progress from sensor integration and data collection to model development, deployment, and experimental evaluation. The resulting laboratory-scale prototype is expected to demonstrate the potential of lightweight Edge AI for local anomaly detection and condition-based maintenance support, providing a foundation for further development and industrial validation.

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